CHANGELOOP

Video Script

A screen-recording narration for the SANKALP 2026 submission — the problem, why it has not been solved, and how ChangeLoop solves it, step by step.

What you do: screen-record the running website and read this out.

Two complete versions, for two different moments:

Version	Runs	Use it for
B — Video	**~3:24**	**the submission video. Record this one.**
A — Live pitch	**~8:30**	if you are shortlisted and present live to the jury — the full story, with every reason spelled out

B tells the same story as A, in the same order, with the same clicks — just shorter. Learn B for the video now. Read A before any live round: it is the version you will be speaking from when a judge says "tell us more".

Everything in a **grey box is what you say.** Everything in **bold outside a box is what you click.** Lines starting **Need:** are not spoken — they remind you why each step exists, so you can answer if a judge asks.

What the jury must think, in order

By the end of each part, the jury should be thinking one sentence:

Part	What the jury should be thinking
The river	*"Real people were harmed by this."*
Today	*"This is a big problem — and it is still getting worse."*
One morning	*"I can see exactly how it happens, every single day."*
Why nobody fixes it	*"Nobody is even looking in the right place."*
Why it cannot wait	*"This has to be solved."*
The insight	*"I did not know that. Nobody told me that."*
Steps 1–5	*"He has actually solved it — and I can see how."*
The ending	*"This is the right answer."*

Salt is the thread through all of it:

```
salt poisoned the farmers' wells
   → the court said: release nothing
filters clean the water — but cannot get rid of salt
   → so it is boiled out, and boiling burns fuel
and then the salt is still there — over a lakh tonnes, in sheds, with nowhere to go
   → everyone is trying to get rid of it at the end
but the amount is decided at the start — at six in the morning, by one planner
   → ChangeLoop works at that desk
```

Set up before recording

```
cd changeloop
python backend/server.py
```

Open `http://localhost:8000`

- Click **Reset**
- **Binding constraint → Normal operation**
- Full screen (F11), zoom 100%, close every other tab
- Put this script on a second screen or your phone

Record on your own computer, not the Render link. Render sleeps and takes 50 seconds to wake.

The flow picture does not move at the start. That is correct. It starts moving when you click **Accept**.

Speak slowly. The pauses are doing work.

VERSION A

LIVE PITCH (~8:30)

A1. The river

On screen: the Command page. Do not touch the mouse. Let them look.

SAY THIS

In 2003, farmers in Tamil Nadu took an industry to court.

Not an NGO. Farmers.

Their river is the Noyyal. Upstream is Tirupur — the town that makes more than half of India's knitwear exports.

To dye cotton you need two things: colour, and a lot of salt. The salt pushes the dye into the fibre. For years, both went into that river.

(pause)

SAY THIS

You could see the colour. You could not see the salt — and salt stays.

The groundwater turned salty. Wells that had fed that land for generations became useless.

In 2011, the court shut down seven hundred dyeing units, and gave one order. Release nothing.

A2. Today — the fuel, and the mountain

SAY THIS

Tirupur rebuilt. Four hundred and fifty dyeing units spent over a thousand crore rupees on shared treatment plants. Today they recycle a hundred and thirty million litres of water every day.

It worked. The river is cleaner.

(pause — slow down, this is the step everything rests on)

SAY THIS

But filters do not get rid of salt. They only squeeze it into less and less water — until you are holding a small amount of extremely salty water that you are not allowed to pour anywhere.

So you boil it, until only dry salt is left. Boiling needs steam. Steam comes from a boiler, mostly burning wood.

The salt that once poisoned the river now burns fuel.

(pause)

SAY THIS

And when the boiling is done — the salt is still there.

Last year, the Dyers Association of Tiruppur said more than one lakh tonnes of it is sitting in sheds at the treatment plants. Three years earlier, reports put it at fifty to sixty thousand. It has nearly doubled.

In their own words: "We don't have the technology for this salt, so we pile up huge amounts of salt."

So every dyehouse pays for its salt twice. Once to buy it. And once to boil it back out. And then nobody knows where to put it.

A3. One morning in one dyehouse

Still on Command.

SAY THIS

Here is how that happens, on an ordinary morning.

It is six o'clock. Five orders came in overnight. Deep indigo. Pale mint. Dusty rose. Carbon black. And natural ecru — almost white.

The planner runs them in the order they arrived.

So after the carbon black, the machine has to be scrubbed clean before the ecru goes in — or the white cloth comes out grey. That one clean-out uses over seven thousand litres of water, and every litre carries salt to the treatment plant.

The washing runs on a fixed timer, to be safe — even after the cloth is already clean.

And by two in the afternoon, this one machine has used up its share of the factory's water allowance.

(pause)

SAY THIS

Nobody did anything wrong. They followed the system they were given.

And this happens every morning, in every one of those four hundred and fifty units.

A4. Why nobody has fixed it

SAY THIS

So why has nobody solved this? Three reasons.

First — everyone is working at the end of the pipe. Better filters. Better boilers. Some way to get rid of the salt. The Dyers Association says that technology is "yet to evolve." Nobody is working on the start — on how much salt goes in.

Second — the person who decides that is the planner, at six in the morning. And the planner never sees the bill. It arrives weeks later, at a shared plant, split across four hundred and fifty factories.

Third — the tools that exist count water. And that is exactly what almost everyone gets backwards.

A5. Why it cannot wait

> **SAY THIS**
>
> And this cannot wait.
>
> That salt cannot be released. There is nowhere to send it. And it grows by up to seventy tonnes, every single day.
>
> And the same rule — release nothing — now applies across India. To tanneries. To distilleries. To paper mills. Every one of them will meet the same salt, and the same fuel bill.

A6. The insight

Move the mouse to the band at the top. Read the big sentence:

> **SAY THIS**
>
> Here is what I found.
>
> In a zero discharge dyehouse, the evaporator's fuel bill is set by how much salt goes into the dye bath. Not by how much water comes out.

Explain:

> **SAY THIS**
>
> The filters can only make that leftover water so salty before they clog. So how much you boil depends on the salt. Use more water with the same salt — and you still boil the same amount.

Point at the four numbers on the right of the band.

> **SAY THIS**
>
> Cut the water by twenty percent: the energy changes by zero point zero percent.
>
> Cut the salt by twenty percent: it drops by twenty.
>
> Water is the carrier. Salt is the load. And this is ChangeLoop — the system I built to act on it. Let me show you, step by step, the same morning — fixed.

A7. Step one — the six o'clock plan

Why this step exists: the salt is decided at the planner's desk, so the solution has to work there.

Click Run optimiser. Then Decisions.

> **SAY THIS**
>
> Step one. ChangeLoop starts where the problem starts — the planner's desk, at six in the morning.
>
> It takes those same five orders and checks four hundred and eighty ways to run them.
>
> Its answer: run the ecru first, while the machine is still clean. That one change removes the seven thousand litre clean-out.
>
> It cannot simply sort them light to dark — the indigo is due by ten. So it finds the order that saves the most and still keeps every deadline.

Move the mouse to Option C. Slow down.

> **SAY THIS**
>
> And look at Option C. It would save almost half the fresh water.
>
> ChangeLoop refuses it. It makes one order two point six hours late — and no factory will use a tool that loses it a customer.

A8. Step two — the planner decides

Why this step exists: no factory will hand control of its machines to software.

Click Accept. *The flow picture starts moving — point at it.*

> **SAY THIS**
>
> Step two. Nothing happens until a person says yes. The planner approves, and it is recorded under their name.
>
> Reject it — and the saving counts as zero.

A9. Step three — stop washing when it is actually clean

Why this step exists: a fixed timer wastes water and salt — but stopping too early ruins the batch.

Click Water. Scroll to Inject a fault.

> **SAY THIS**
>
> Step three — the washing. Instead of a fixed timer, ChangeLoop watches the sensors and tells the operator when the cloth is truly clean.
>
> But stop too early and the colour bleeds, and the whole batch is washed again. So the rule is strict.

Click Calibration drift. Click Attempt release.

> **SAY THIS**
>
> When a sensor cannot be trusted — refused. At the gate, and again at the server. It never releases a batch on its own.

Stay silent for two seconds.

Click No fault. Then Grant release.

A10. Step four — no more surprise at two o'clock

Why this step exists: a factory must know it will break its water allowance *before* it happens.

Click Forecast.

> **SAY THIS**
>
> Step four — the two o'clock problem. ChangeLoop shows it at six in the morning, while there is still time to act.
>
> Doing nothing: a hundred and thirty-two percent of the allowance.
>
> Reusing the rinse water — the fix everyone pays for: still a hundred and thirty-two.

(pause)

> **SAY THIS**
>
> The same. Only cutting the salt brings it inside — at eighty-two.

A11. Step five — numbers that hold up

Why this step exists: under a court order, every number must survive an audit.

Click Evidence.

> **SAY THIS**
>
> Step five. Under a court order, every number has to stand up. So I will say this first: nothing here is measured yet.
>
> Seventeen numbers come from published sources. Four are calculated. Four are assumed.
>
> But the model can be tested. The Pollution Control Board measured the salt in a real Tirupur unit's water. Fed that number, my model predicts thirty point six percent — inside the twenty to thirty percent real plants report.

A12. The ending

Click Impact.

> **SAY THIS**
>
> On one machine, in one shift: sixty kilos less salt sent to the treatment plant. Nearly ten thousand rupees. Every delivery on time.
>
> And a dyehouse runs many machines. Every shift. Every day.

(pause)

> **SAY THIS**
>
> Remember that shed. More than a lakh tonnes of salt — and the industry is still searching for a way to get rid of it.
>
> But every kilo in that shed started in a dye bath. Decided at six in the morning, by a planner who never saw the bill.

(pause — slow down for the last three lines)

> **SAY THIS**
>
> Getting rid of the salt is one answer.
>
> Not putting it in is a better one.
>
> That is the choice we change.

Hold still for three seconds. Then end the recording.

VERSION B

VIDEO (~3:43)

This is the one to record. Same story, same order, same clicks as A — shorter sentences. Read it straight through, at your normal pace between the marked pauses. The greeting adds about twenty seconds versus the old 3:24 cut — if your limit is a hard 3:30, drop B0 and start straight at B1.

B0. The greeting — *Command page, mouse still, before anything else*

SAY THIS

Good morning everyone. My name is Shorya Ashish Mittal, and I am going to show you something that surprised me while I was building this project.

Before I explain how my system works, let me tell you why it needed to exist in the first place.

B1. The river — *same screen, continue*

SAY THIS

In 2003, farmers in Tamil Nadu took an industry to court.

Upstream of their river sits Tirupur, which makes more than half of India's knitwear exports. Dyeing cotton needs colour and a lot of salt — and for years both went into that river. The wells turned salty.

In 2011 the court shut down seven hundred dyeing units, and ordered: release nothing.

B2. Today

SAY THIS

Tirupur rebuilt, and now recycles a hundred and thirty million litres a day.

But filters do not get rid of salt. They squeeze it into less water, until it has to be boiled dry — with steam from the boiler.

And then the salt is still there. The Dyers Association says more than one lakh tonnes now sits in sheds — nearly double what it was three years ago. In their words: "We don't have the technology for this salt."

B3. One morning

> **SAY THIS**
>
> On an ordinary morning, five orders arrive — from deep indigo to almost-white ecru. Run in the order they came, the machine must be scrubbed after carbon black before the ecru: seven thousand litres of cleaning. Washing runs on a fixed timer. By two in the afternoon, the machine is over its water limit — in every one of four hundred and fifty units.

B4. Why nobody fixes it — and why it cannot wait

> **SAY THIS**
>
> Nobody fixes this because everyone works at the end of the pipe. The planner who decides the salt never sees the bill. And today's tools count water.
>
> And it cannot wait: the salt grows by seventy tonnes a day, and the same rule now covers tanneries, distilleries and paper mills across India.

B5. The insight — *read the big sentence, then point at the numbers*

> **SAY THIS**
>
> Here is what I found. In a zero discharge dyehouse, the evaporator's fuel bill is set by how much salt goes into the dye bath. Not by how much water comes out.
>
> Cut water twenty percent: energy moves zero point zero. Cut salt twenty percent: it drops twenty. This is ChangeLoop. Step by step:

B6. Step one — Run optimiser → Decisions

> **SAY THIS**
>
> One: at the planner's desk, it checks four hundred and eighty plans — run the ecru first, while the machine is clean, and keep every deadline. Option C saves more water but makes an order late, so it is refused.

B7. Step two — Accept

> **SAY THIS**
>
> Two: a person approves, and it is recorded.

B8. Step three — Water → Calibration drift → Attempt release

> **SAY THIS**
>
> Three: washing stops when the sensors say the cloth is clean — and when a sensor cannot be trusted, release is refused.

No fault → Grant release.

B9. Step four — Forecast

> **SAY THIS**
>
> Four: the two o'clock problem is visible at six. Reusing rinse water: still a hundred and thirty-two percent. Only cutting salt brings it inside.

B10. Step five — Evidence

> **SAY THIS**
>
> Five: nothing here is measured yet — I say that first. But fed real plant data, the model predicts what real plants report.

B11. The ending — Impact

> **SAY THIS**
>
> One machine, one shift: sixty kilos less salt, nearly ten thousand rupees, every delivery on time.
>
> Every kilo in that shed started in a dye bath — decided at six in the morning, by a planner who never saw the bill.
>
> Getting rid of the salt is one answer. Not putting it in is a better one.
>
> That is the choice we change.

Hold still for three seconds. Then end the recording.

REFERENCE

Numbers to get right

Keep this open in another window.

Where	Number	Source
Units shut in 2011	**700**	Down To Earth; Economic Times
Tirupur's share of India's knitwear exports	**more than half** (54%)	Economic Times, June 2025
Dyeing units / shared plants	**450 / 18**	Economic Times, June 2025
Spent on treatment plants	**over ₹1,000 crore** (₹1,013 cr)	Economic Times, June 2025
Recycled today	**130 million litres a day**	Economic Times, June 2025
Salt piled in sheds, 2022	**50,000 to 60,000 tonnes**	DT Next and Ecotextile News, Oct 2022
Salt piled in sheds, 2025	**over 1 lakh tonnes**	Dyers Association, via Economic Times, June 2025
Salt added per day	**up to 70 tonnes**	DT Next, Oct 2022
Clean-out after carbon black	**over 7,000 L** (7,200 L)	the engine
Indigo due	**by 10 am**	the engine
Cut water 20% → energy	**0.0%**	the engine
Cut salt 20% → energy	**−20.0%**	the engine
Plans checked	**480**	the engine
Option C late by	**2.6 hours**	the engine
Do nothing / reuse rinse / low salt	**132.3% / 132.3% / 82.1%**	the engine
Model predicts at CPCB inlet	**30.6%** vs published **20–30%**	the engine; CPCB
Salt avoided, one machine, one shift	**about 60 kg** (62.6 kg)	the engine
Cost avoided, same	**nearly ₹10,000** (₹9,709)	the engine

If the screen shows something different, the screen is correct.

If a judge asks

Is the one lakh tonnes real? Yes. The Dyers Association of Tiruppur told the Economic Times in June 2025 that accumulated mixed waste salt at the treatment plants exceeds 100,000 tonnes, stored in sheds because no disposal method has been found. DT Next reported nearly 50,000 tonnes in October 2022, growing by up to 70 tonnes a day.

Is the six o'clock morning a real factory? It is the exact queue the engine runs — five real-shaped orders with real shade depths and delivery times. Every number in that story is what the screen shows. It is a worked example, not a named customer, and I would say so.

Who is the customer? The dyehouse owner. They buy the salt, and they pay the treatment plant to boil it out.

Who would sell it? The shared treatment plant. Every kilo of salt its members do not send is steam it does not buy and salt it does not have to store. Its incentive already points the same way as ours.

"Pays for its salt twice" — true? Yes. Once to buy it for the dye bath, once through the treatment charge to boil it out. Both are separate costs in the model and never added together into the water figure.

Why "many machines" rather than a bigger number? Because a cluster-wide total would be a projection from one modelled machine. "Many machines, every shift, every day" is true without claiming a figure we have not measured.

What is the weakness? We have not measured a real site yet. One visit and two instrument readings — steam flow to the evaporator and the saltiness of the leftover water — would turn this from a checked model into a measured one. That is the first thing we would do with support.

Words to avoid

Do not say	Say instead
"AI", "machine learning"	"a search that checks every option"
"blockchain"	"a tamper-evident record"
"we saved", "our result"	"the model shows"
"we measured"	"we checked it against published plant data"
any jobs figure for 2011	"tens of thousands"
"a factory we worked with"	"a typical morning" — it is a worked example

If something goes wrong

- Small mistake — keep going. Calm recovery beats a visible cut.
- Wrong page — go back and say "let me show you that properly".
- Wrong state — click **Reset**, restart from Command.
- Flow not moving — correct before **Accept**.
- A number looks odd — say what is on screen.

The five moments that must survive

Whichever version you use, never lose these:

1. **"The salt that once poisoned the river now burns fuel."**
2. **The shed — over a lakh tonnes, nearly doubled in three years.** The real, current problem.
3. **"Zero point zero percent."** The insight.
4. **Option C refused, and the gate refusing your own release.** The proof it can be trusted.
5. **"Getting rid of the salt is one answer. Not putting it in is a better one."**

One last thing

Every other team will say "we save water." You are the one who will show the jury a shed with a lakh tonnes of salt in it, explain why saving water does not shrink it, and then show the exact desk, at the exact hour, where it can be shrunk — without ever missing a delivery.

Do not rush the first two minutes. The story is what makes the software matter.